

Future Methods



Agenda



- What are Future Methods
- Why do we need Future Methods
- Basic Syntax and Considerations
- Lets write our first Future Method

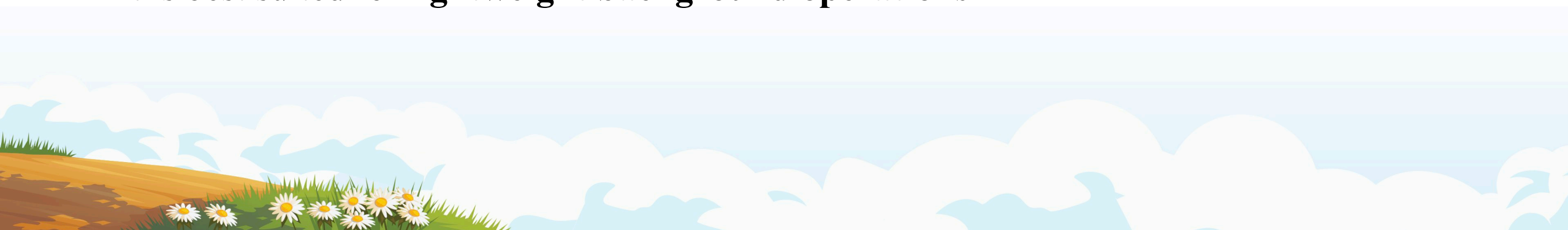


What are Future Methods :

A Future Method is a type of Asynchronous Apex that executes in the background after the main transaction is completed. It runs in a separate Apex thread and is commonly used for non-urgent tasks that should not block the user or slow down the current process.

Key Characteristics of a Future Method:

- It executes **after the current transaction commits**
- It always runs in a **separate Apex transaction**
- The **user does not have to wait** for its execution
- It is best suited for **lightweight background operations**

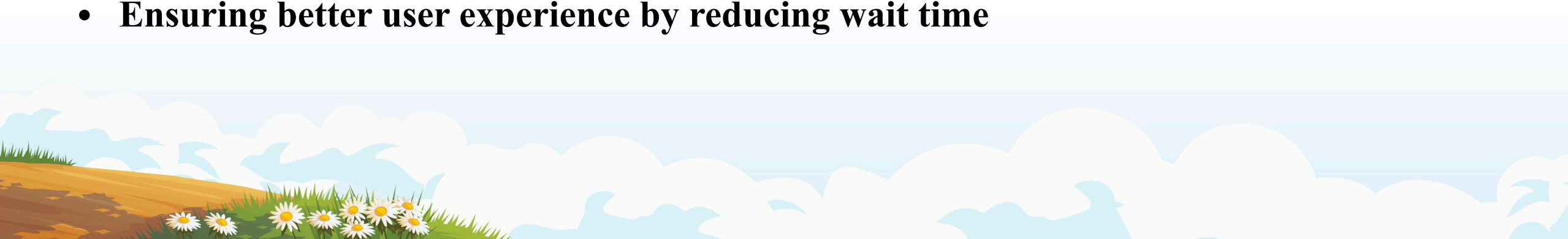


Why do we need Future Methods:

Future Methods are required in Salesforce to solve practical real-world limitations of synchronous processing.

They are commonly used for:

- **Making HTTP callouts from triggers**, which is not allowed synchronously
- **Handling Mixed DML issues**, where setup and non-setup objects are involved
- **Offloading heavy post-processing work** from the main transaction
- **Improving overall system performance**
- **Ensuring better user experience by reducing wait time**



Future Method Syntax and Considerations :



- A Future Method must always be declared as **static**
- A Future Method must **return void** and cannot return any value
- A Future Method **cannot accept sObjects directly** as parameters
- It can accept only **primitive data types, list of primitives, or map of primitives**
- A Future Method **cannot call another Future Method**
- A Future Method **cannot call a Queueable job**
- The **execution order of Future Methods is not guaranteed** by Salesforce
- Future Methods **cannot be monitored or tracked** like Queueable Apex
- Future Methods are best suited only for **simple and lightweight background processing**





Lets Write our first future Method

Use case - Create a follow up task every time a new Lead gets created and associate that task with Lead.



thank you

